



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Securitization

1. Grow More Ltd. an NBFC is in the need of funds and hence it sold its receivables to MAC Financial Corporation (MFC) for ₹ 100 million. MFC created a trust for this purpose called General Investment Trust (GIT) through which it issued securities carrying a different level of risk and return to the investors. Further, this structure also permits the GIT to reinvest surplus funds for short term as per their requirement.

MFC also appointed a third party, Safeguard Pvt. Ltd. (SPL) to collect the payment due from obligor(s) and passes it to GIT. It will also follow up with defaulting obligor and if required initiate appropriate legal action against them.

From the information given above, choose the correct answer to the following questions:

- I. The securitized instrument issued for ₹ 100 million by the GIT falls under category of
 - (a) Pass Through certificate (PTCs)
 - (b) Pay Through Security (PTS)
 - (c) Stripped Security
 - (d) Debt Fund.
- II. In the above scenario, the Originator is.....
 - (a) Grow More Ltd.

- (b) MAC Financial Corporation (MFC)
 - (c) General Investment Trust (GIT)
 - (d) Safeguard Pvt. Ltd.
- III. In the above scenario, the General Investment Trust (GIT) is a/an.....
- (a) Obligor
 - (b) Originator
 - (c) Special Purpose Vehicle (SPV)
 - (d) Receiving and Paying Agent (RPA)
- IV. In the above scenario, the Safeguard Pvt. Ltd. (SPL) is a/an.....
- (a) Obligor
 - (b) Originator
 - (c) Special Purpose Vehicle (SPV)
 - (d) Receiving and Paying Agent (RPA)
- V. Which of the following statement holds true?
- (a) When Yield to Maturity in market rises, prices of Principle Only (PO) Securities tend to rise.
 - (b) When Yield to Maturity in market rises, prices of Principle Only (PO) Securities tend to fall.
 - (c) When Yield to Maturity in market falls, prices of Principle Only (PO) Securities tend to fall.
 - (d) When Yield to Maturity in market falls, prices of Principle Only (PO) Securities remain the same.

International Financial Management

2. A US parent company has subsidiaries in France, Germany, UK and Italy. The amounts due to and from the affiliates is converted into a common currency viz. US dollar and entered in the following matrix.

Inter Subsidiary Payments Matrix

(US \$ Thousands)

		Paying affiliate				
		France	Germany	UK	Italy	Total
Receiving affiliate	France	---	80	120	200	400
	Germany	120	---	80	160	360
	UK	160	120	---	140	420
	Italy	200	60	120	---	380
	Total	480	260	320	500	1560

The treasurer of US Parent company is suggesting that by applying Multilateral Netting system the company can save a lot of transfer/exchange costs. The company's Board agreed with Treasurer's proposal.

From the information given above, choose the correct answer to the following questions:

- I. Before applying Multilateral Netting it is necessary to apply.....
 - (a) Unilateral Netting
 - (b) Bilateral Netting
 - (c) Multilateral Netting
 - (d) Interest Rate Swapping
- II. Through Multinational Netting these transfers will be reduced to
 - (a) \$ 50,000
 - (b) \$ 100,000
 - (c) \$ 150,000
 - (d) \$ 200,000
- III. The Net Payment/ Net Receipts for France after netting off shall be.....
 - (a) Net Receipt \$ 40,000
 - (b) Net Payment \$ 80,000

- (c) Net Payment \$ 40,000
- (d) Net Receipt \$ 80,000
- IV. The Net Payment/ Net Receipts for Italy after netting off shall be.....
 - (a) Net Receipt \$ 60,000
 - (b) Net Payment \$ 120,000
 - (c) Net Payment \$ 60,000
 - (d) Net Receipt \$ 120,000
- V. Suppose if the transfer charges are 0.01% of the amount transferred then by applying multilateral netting techniques there will be reduction in overall cost of transfer by
 - (a) US \$ 136
 - (b) US \$ 156
 - (c) US \$ 1,360
 - (d) US \$ 1,560

Advanced Capital Budgeting Decisions

3. SS Company is considering the replacement of its existing machine with a new machine. The Purchase price of the new machine is ₹ 26 Lakhs and its expected Life is 8 years. The company follows straight-line method of depreciation on the original investment (scrap value is not considered for the purpose of depreciation). The other expenses to be incurred for the New Machine are as under:
- (i) Installation Charges ₹ 9,000
 - (ii) Fees paid to the consultant for his advice to buy New Machine ₹ 6,000.
 - (iii) Additional Working Capital required ₹ 17,000. (will be released after 8 years)

The written down value of the existing machine is ₹ 76,000, and its Cash Salvage Value is ₹ 12,500. The dismantling of this machine would cost

₹ 4,500. The Annual Earnings (before tax but after depreciation) from the New Machine would amount to ₹ 3,15,000. Income tax rate is 35%. The Company's required Rate of Return is 13%.

You are required to advise on the viability of the proposal.

$$PVIF (13\%, 8) = 0.376 \quad PVIFA (13\%, 8) = 4.800$$

Security Valuation

4. R Ltd. has surplus cash of ₹ 100 lakhs and wants to distribute 27% of it to the shareholders. The company decides to buy back shares. The Finance Manager of the company estimates that its share price after re-purchase is likely to be 10% above the buyback price if the buyback route is taken. The number of shares outstanding at present is 10 lakhs and the current EPS is ₹ 3.

You are required to suggest:

- The price at which the shares can be re-purchased, if the market capitalization of the company should be ₹ 210 lakhs after buyback,
 - The number of shares that can be re-purchased, and
 - The impact of share re-purchase on the EPS, assuming that after buyback net income shall remain the same.
5. A hypothetical company ABC Ltd. issued a 10% Debenture (Face Value of ₹ 1000) of the duration of 10 years is currently trading at ₹ 850 per debenture. The bond is convertible into 50 equity shares being currently quoted at ₹ 17 per share.

If yield on equivalent comparable bond is 11.80%, then determine the spread of yield of the above bond from this comparable bond.

The relevant present value table is as follows.

Present Values	t ₁	t ₂	t ₃	t ₄	t ₅	t ₆	t ₇	t ₈	t ₉	t ₁₀
PVIF _{0.11, t}	0.901	0.812	0.731	0.659	0.593	0.535	0.482	0.434	0.391	0.352
PVIF _{0.13 t}	0.885	0.783	0.693	0.613	0.543	0.480	0.425	0.376	0.333	0.295

Business Valuation

6. STR Ltd.'s current financial year's income statement reported its net income after tax as ₹ 50 Crore.

Following is the capital structure of STR Ltd. at the end of current financial year:

	₹
Debt (Coupon rate = 11%)	80 Crore
Equity (Share Capital + Reserves & Surplus)	250 Crore
Invested Capital	330 Crore

Following data is given to estimate cost of equity capital:

Asset Beta of TSR Ltd.	1.11
Risk –free Rate of Return	8.5%
Average market risk premium	9%

The applicable corporate income tax rate is 30%.

Estimate Economic Value Added (EVA) of RST Ltd. in ₹ lakh.

Derivatives Analysis & Valuation

7. (i) On 1 April 2015, Sunidhi was holding a portfolio of 10 securities whose value was ₹ 9,94,450, the weighted average of beta of 9 securities of the portfolio was 1.10.

Since she was expecting a fall in the prices of the shares in near future to hedge her portfolio, she sold 5 contract of NIFTY Futures (Multiplier of 25) expiring in May 2015, which was trading at 8767.07 on 1 April.

Required:

- (1) Calculate the beta of the 10th security.
- (2) Reconcile the reasons in spite of 2% fall in the market as per Sunidhi's apprehension if she would have earned some profit on her cash position.

- (ii) A Futures contract is available on R Ltd. that pays an annual dividend of ₹ 4 and whose stock is currently priced at ₹ 125. Each Futures contract calls for delivery of 1,000 shares to stock in one year, daily marking to market. The treasury bill rate is 8%.

Required:

- (1) Given the above information, assess the price of one Futures contract.
- (2) If the company stock price decreases by 6%, then what will be the price of one Futures contract?
- (3) Suppose the company stock price decreases, then evaluate realizes gain or loss for an investor that has a long position in one Futures contract of R Ltd.

(Ignore margin and taxation, if any)

Interest Rate Risk Management

8. MPD Ltd. issues a ₹ 50 Million Floating Rate Loan on July 1, 2018 with resetting of coupon rate every 6 Months equal to LIBOR + 50 bps.

MPD is interested in an Interest rate Collar Strategy of selling a Floor and buying a cap.

MPD buys the 3 years cap and sell 3 years Floor as per the following details on July 1, 2018:

Principal Amount	₹ 50 Million
Strike Rate	5% for Floor & 8% for Cap
Reference Rate	6 months LIBOR
Premium	NIL, since premium paid for cap = premium received for Floor

The Reset dates & Interest rates p.a., on that dates are:

Reset Date	31/12/2018	30/06/2019	31/12/2019	30/06/2020	31/12/2020	30/06/2021
LIBOR (%)	7.00	8.00	6.00	4.75	4.25	5.25

Using the above data, you are required to determine:

- (i) Effective Interest paid out at each six reset dates, (Round off to the nearest rupee)
- (ii) Average overall effective rate of interest p.a. (round off to 2 decimals).

Note: Assume 365 days a year and 366 days for leap year.

Portfolio Management

9. Mr. NK has categorized all the available stock in the market into the following types:

- (i) Small cap growth stocks
- (ii) Small cap value stocks
- (iii) Large cap growth stocks
- (iv) Large cap value stocks

Mr. NK also estimated the weights of the above categories of stocks in the market index. Further, the sensitivity of returns on these categories of stocks to the three important factors are estimated to be:

Category of Stocks	Weight in the Market Index	Factor I (Beta)	Factor II (Book Price)	Factor III (Inflation)
Small cap growth	25%	0.80	1.39	1.35
Small cap value	10%	0.90	0.75	1.25
Large cap growth	50%	1.165	2.75	8.65
Large cap value	15%	0.85	2.05	6.75
Risk Premium		6.85%	-3.5%	0.65%

The rate of return on Treasury Bonds is 4.50%.

Required:

- (i) Using Arbitrage Pricing Theory, determine the expected return on the market index.

- (ii) Using Capital Asset Pricing Model (CAPM), determine the expected return on the market index.
- (iii) Mr. NK wants to construct a portfolio constituting only the 'Small cap value' and 'Large cap growth' stocks. If the target beta for the desired portfolio is 1, suggest the composition of his portfolio.

Foreign Exchange Exposure and Risk Management

10. On 1st October, 2024 Mr. G, an exporter, enters into a forward contract with a Bank to sell USD 1,00,000 on 31st December 2024 at INR/USD 85.40. However, at the request of the importer, Mr. G received the amount on 30th November, 2024. Mr. G requested the bank take delivery of the remittance on 30th November, 2024 i.e. before due date.

The inter-bank rate on 30th November 2024 was as follows:

Spot INR/USD 85.22-85.27

One Month Premium 10/15

- (i) If bank agrees to take early delivery then what will be net inflow to Mr. G.
- (ii) If Mr. G can deploy these funds in USD and earn a return of 3% per annum, then advise which option will be better: delivery to the bank or investment in USD?

Note:

- a. Assume 365 days in a year.
 - b. Assume the prevailing prime lending rate is 18% p.a. and term deposit rate is 12% p.a.
 - c. Round off calculations upto two decimal points.
11. XP Pharma Ltd., has acquired an export order for ₹ 10 million for formulations from a European company. The Company has also planned to import bulk drugs worth ₹ 5 million from a company in UK. The proceeds of exports will be realized in 3 months from now and the payments for imports will be due after 6 months from now. The invoicing of these exports and imports can be done in any currency i.e.

Dollar, Euro or Pounds sterling at company's choice. The following market quotes are available.

	Spot Rate	Annualised Premium
₹/\$	87.10/87.20	\$ - 7%
₹ /Euro	101.15/101.20	Euro - 6%
₹ /Pound	116.65/116.75	Pound - 5%

Advise XP Pharma Ltd. in which currency invoicing for both import and export should be opted for.

(Exchange Rate including calculations should be upto two decimal places).

Mergers, Acquisitions and Corporate Restructuring

12. X Ltd. is studying the possible acquisition of Y Ltd. by way of merger. The following data are available in respect of both the companies.

Particulars	X Ltd.	Y Ltd.
Market Capitalization (₹)	75,00,000	90,00,000
Gross Profit Ratio	20%	20%
Inventory Turnover Ratio	5 times	4 times
Debtor Turnover Ratio	3 times	5 times
12% Debenture (₹)	10,00,000	-
10% Debenture (₹)	-	14,40,000
No. of Equity Shares	1,00,000	60,000
Operating Expenses	86%	78%
Corporate Tax Rate	30%	30%
Closing Stock (₹)	15,00,000	5,00,000
Debtors (₹)	10,00,000	8,00,000

You are required to calculate:

- (i) Swap ratio based on EPS & MPS respectively as weightage of 40% and 60%.

- (ii) Post Merger EPS
- (iii) Post Merger market price assuming same PE Ratio of X Ltd.
- (iv) Post Merger gain or loss in EPS.

Mutual Funds

13. On 1st April 2024 Fair Return Mutual Fund has the following assets and prices at 4.00 p.m.

Shares	No. of Shares	Market Price Per Share (₹)
A Ltd.	20000	39.40
B Ltd.	100000	965.20
C Ltd.	20000	528.80
D Ltd.	200000	1349.80
E Ltd.	60000	51.80
No. of units of funds		1600000

Please calculate:

- (a) NAV of the Fund on 1st April 2024.
- (b) Assuming that on 1st April 2024, Mr. X, a HNI, send a cheque of ₹ 2 Crore to the Fund and Fund Manager immediately purchases 36000 shares of C Ltd. and balance is held in bank. Then what will be position of fund.
- (c) Now suppose on 2nd April 2024 at 4.00 p.m. the market price of shares is as follows:

Shares	₹
A Ltd.	40.60
B Ltd.	1027.40
C Ltd.	581.60
D Ltd.	1343.80
E Ltd.	88.40

Then what will be new NAV.

Note: Round off all calculations upto two decimal points.

Financial Policy and Corporate Strategy

14. Following Financial data are available for PQR Ltd. for the financial year ending 2023:

	(₹ in lakh)
8% Debentures	125
10% Bonds (2022)	50
Equity Shares (₹ 10 each)	100
Reserves and Surplus	300
Total Assets	600
Assets Turnovers ratio	1.1
Effective interest rate	8%
Effective tax rate	40%
Operating margin	10%
Dividend payout ratio	16.67%
Current market Price of Share	₹ 14
Required rate of return of investors	15%

You are required to:

- (i) Draw income statement for the year.
- (ii) Calculate its sustainable growth rate of earnings.

Theoretical Questions

15. What do you mean by the term 'Pitch Presentation'. List out the points to be kept in mind while preparing a Pitch Presentation.
16. (a) Explain the various challenges to the Efficient Market Theory.
(b) Explain the reasons why conventional methods are not usefully deployed when valuing companies in distress?

OR

Explain the various actions of the Governments of the foreign (host) country by which Country Risk can be identified. Also explain how a company should assess this risk.